



SIMATS ENGINEERING



ASSESSMENT TOOL 2

CONCEPT MAPPING

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Course Code: DSA0201

Course Name: Computer Vision with OpenCV

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QUESTION 1



Smart Photo Editing + Object
Recognition + AR Application

1. A company is developing a smart photo editing + object recognition + AR app.

Create a concept map connecting:

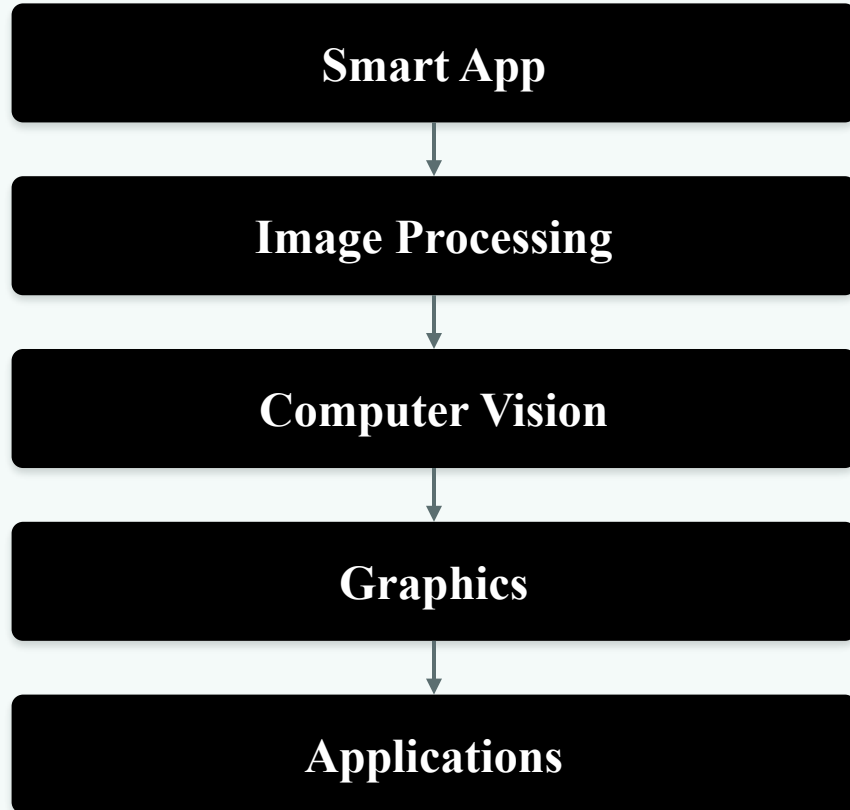
- Image Processing
- Computer Vision
- Computer Graphics
- Levels of Vision
- Applications

Show:

- Workflow of the system
- Interconnections between components
- Real-world examples



Smart App — Flat Chain of Terms



CHARACTERISTICS

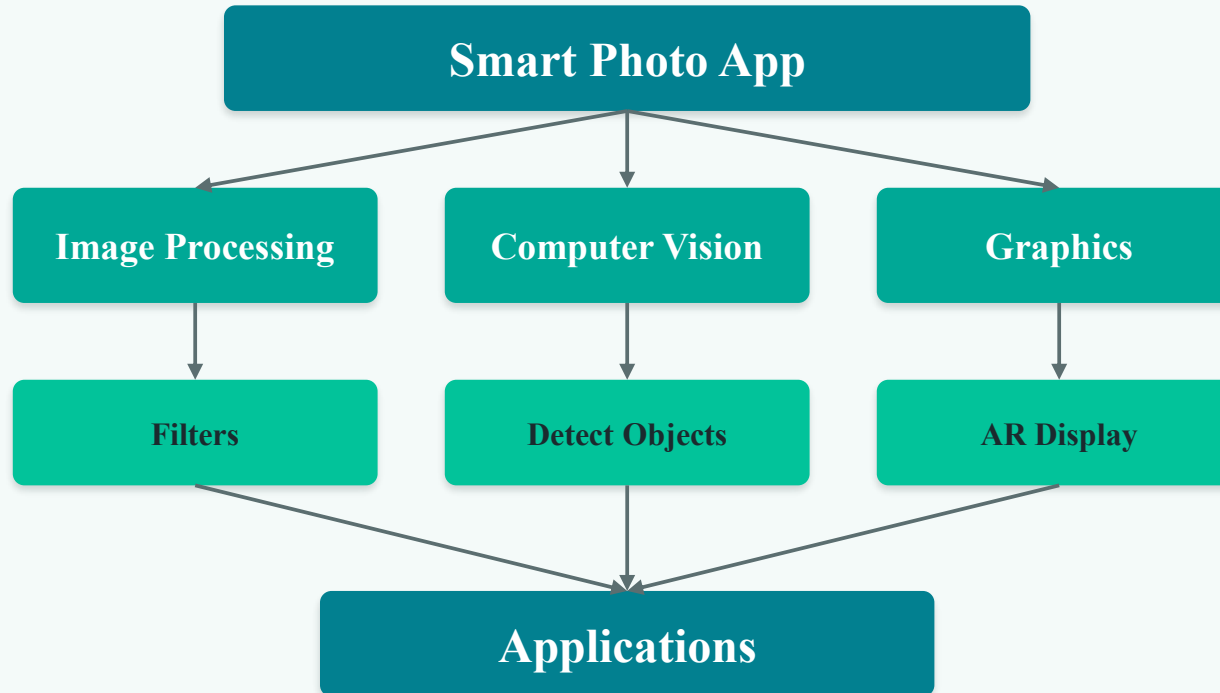
- ✓ Only a handful of concepts listed
- ✓ No workflow or process shown
- ✓ No relationships between ideas
- ✓ No hierarchy — everything is flat
- ✓ No real-world examples given



Level 2 –



Smart Photo App — Basic Branching



CHARACTERISTICS

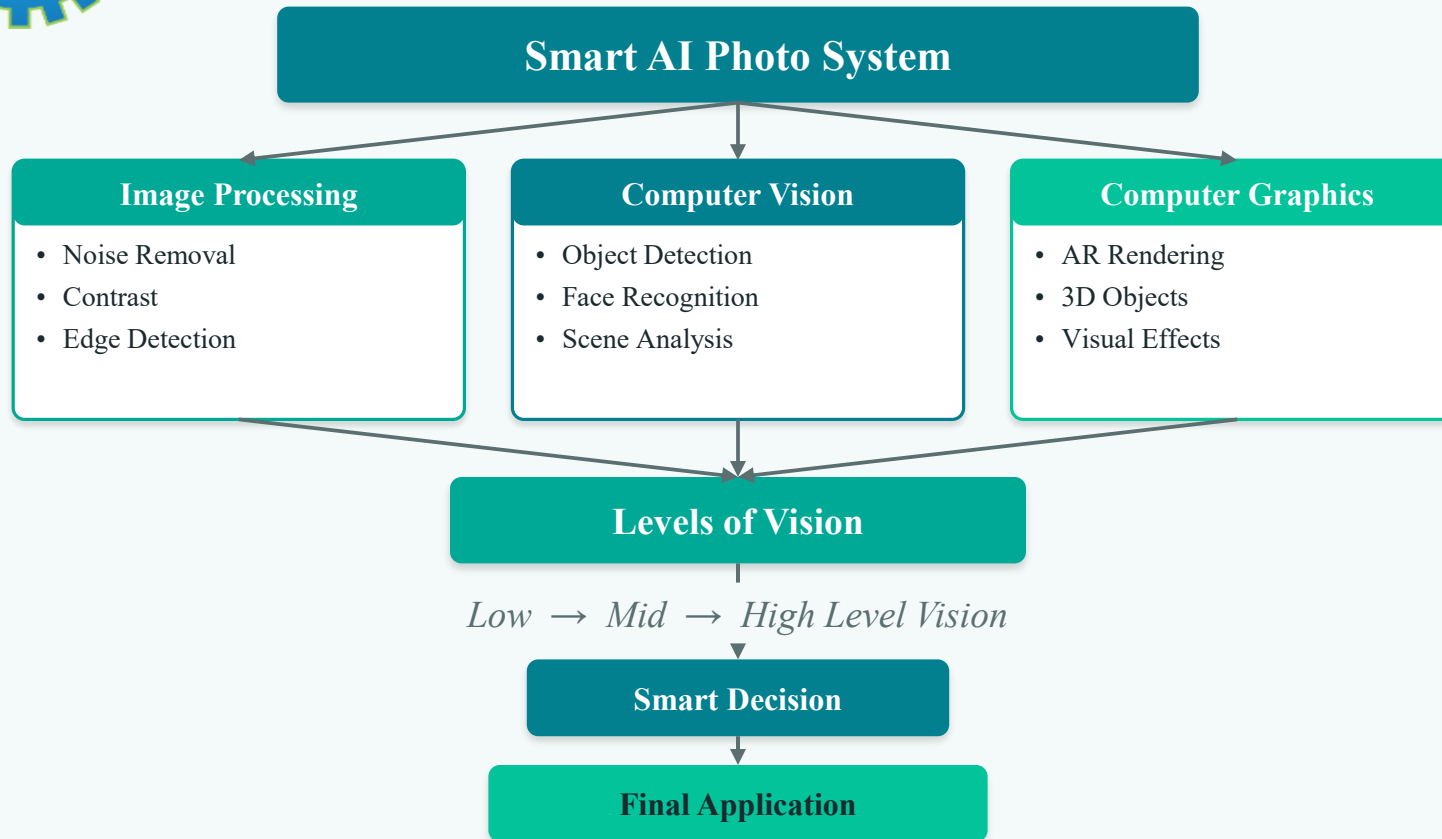
- ✓ Some concepts identified
- ✓ Limited connections shown
- ✓ Weak hierarchy overall
- ✓ Only one application named

Example *Instagram AI Filters*



Smart AI Photo System — Organized Workflow

Level 3
7



CHARACTERISTICS

- ✓ Organized 3-part hierarchy with clear sub-concepts
- ✓ Workflow shown from image to final output
- ✓ Vision levels (low/mid/high) linked to decision-making

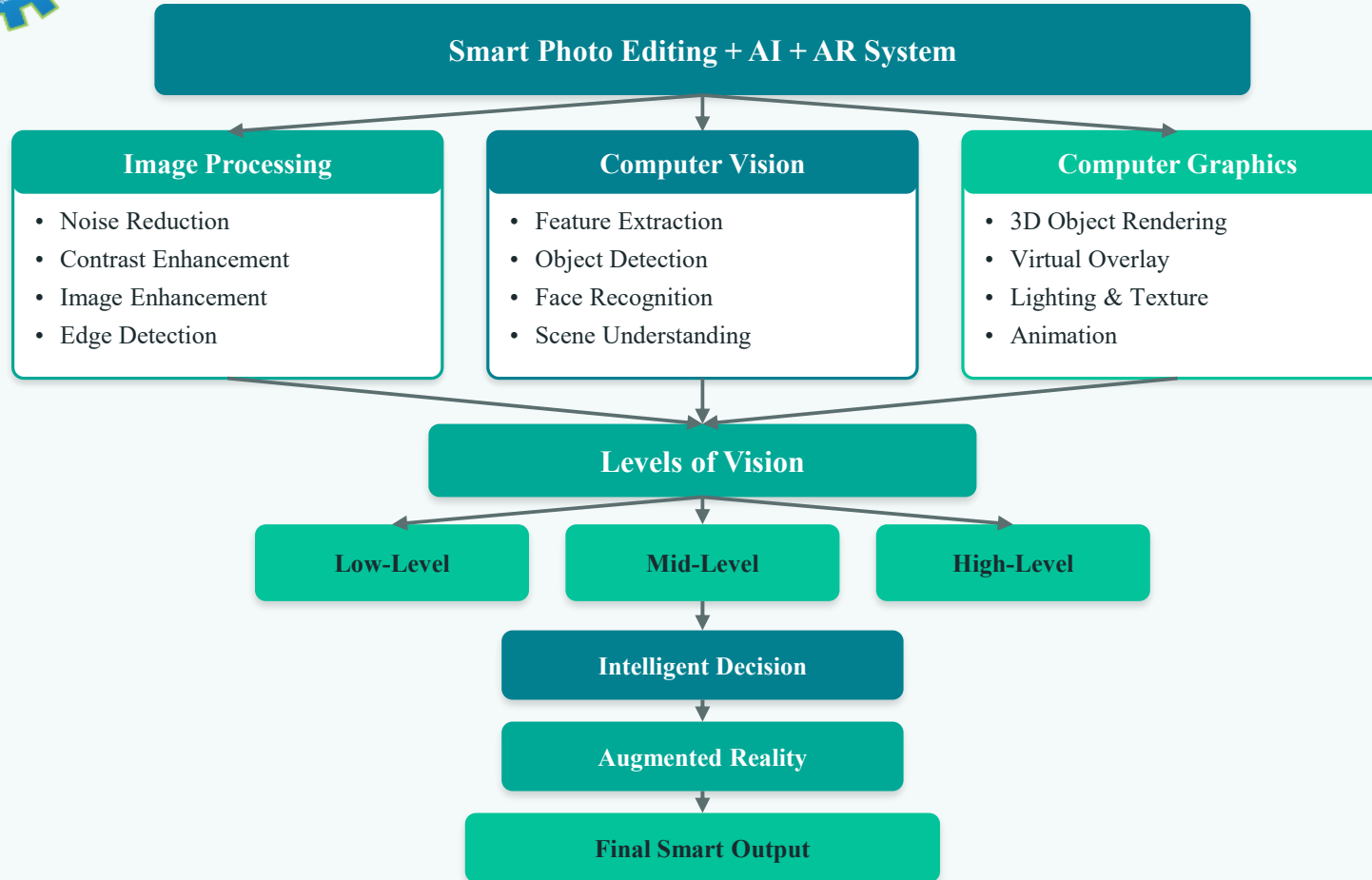
Example *Google Lens, Snapchat, Mobile AR*

WORKFLOW

- Image
- Filtering
- Feature Extraction
- Object Detection
- AR Rendering
- Edited Image



Full System — Integrated Concepts



CHARACTERISTICS

- ✓ All concepts fully identified with rich sub-detail
- ✓ Meaningful linkages across every module
- ✓ Clear hierarchical + sequential organization
- ✓ Concepts integrated end-to-end into one system

APPLICATIONS

- Snapchat Filters
- Google Lens
- IKEA Place AR
- Pokémon GO



QUESTION 2



2. An autonomous drone captures images, calibrates cameras, estimates depth, and reconstructs a 3D scene. Draw a concept map linking:

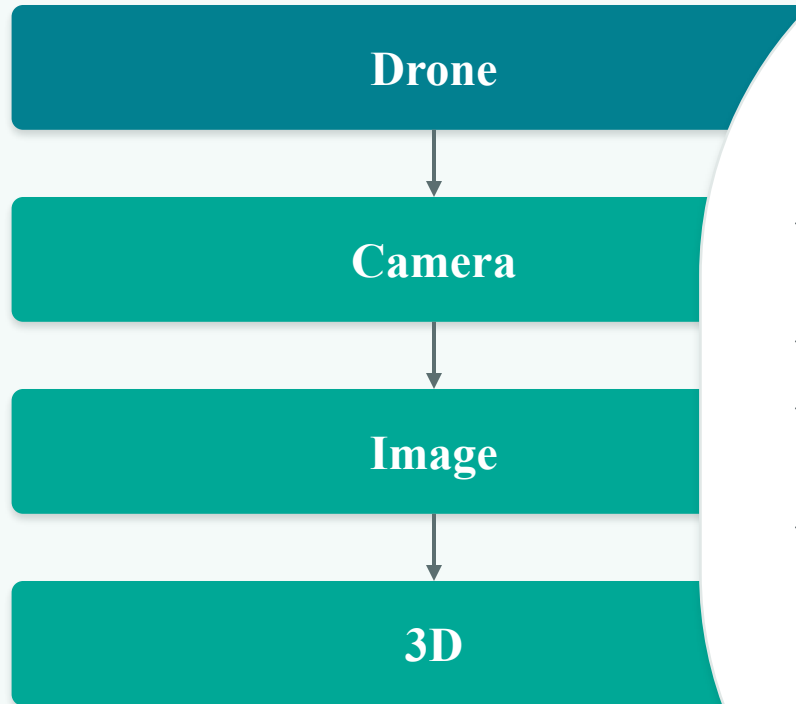
- Camera Model
- Image Formation
- Camera Calibration
- Binocular Vision
- Multiple View Geometry

Include:

- Flow of data
- Dependencies
- Output (3D reconstruction)



Drone — Bare Capture Chain

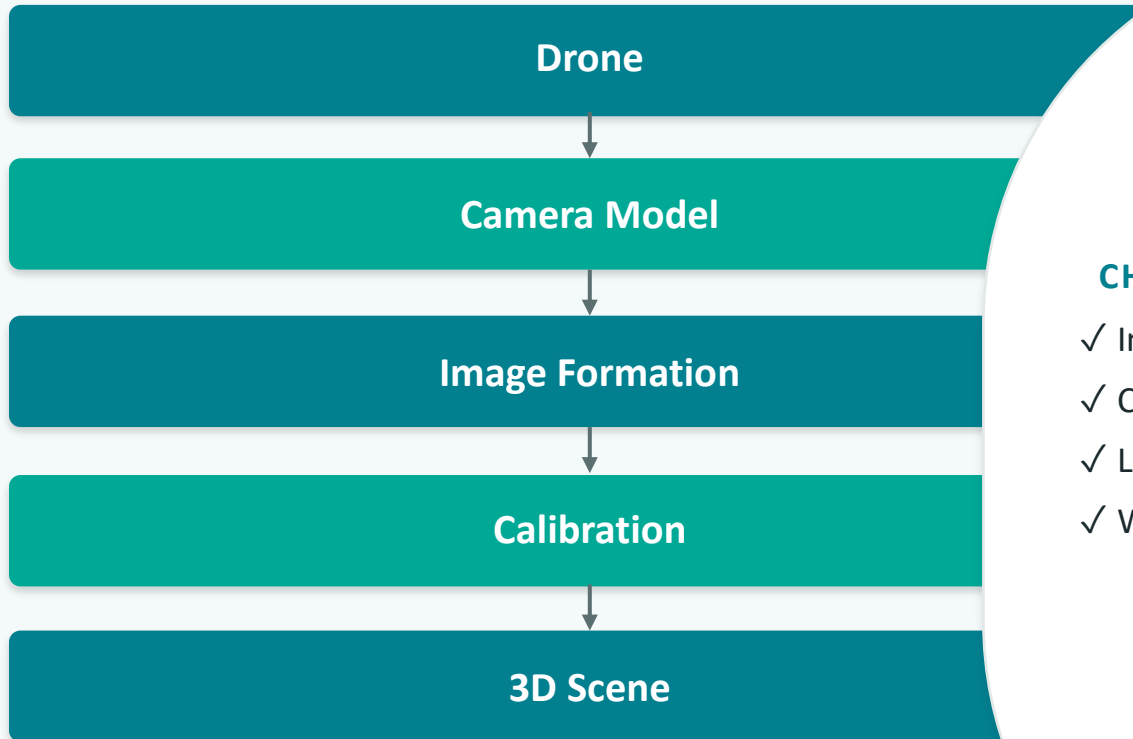


CHARACTERISTICS

- ✓ No dependencies between stages shown
- ✓ No calibration step included
- ✓ No geometry or spatial reasoning mentioned
- ✓ Just a bare list of 4 words



Camera-to-Scene — Weak Linking

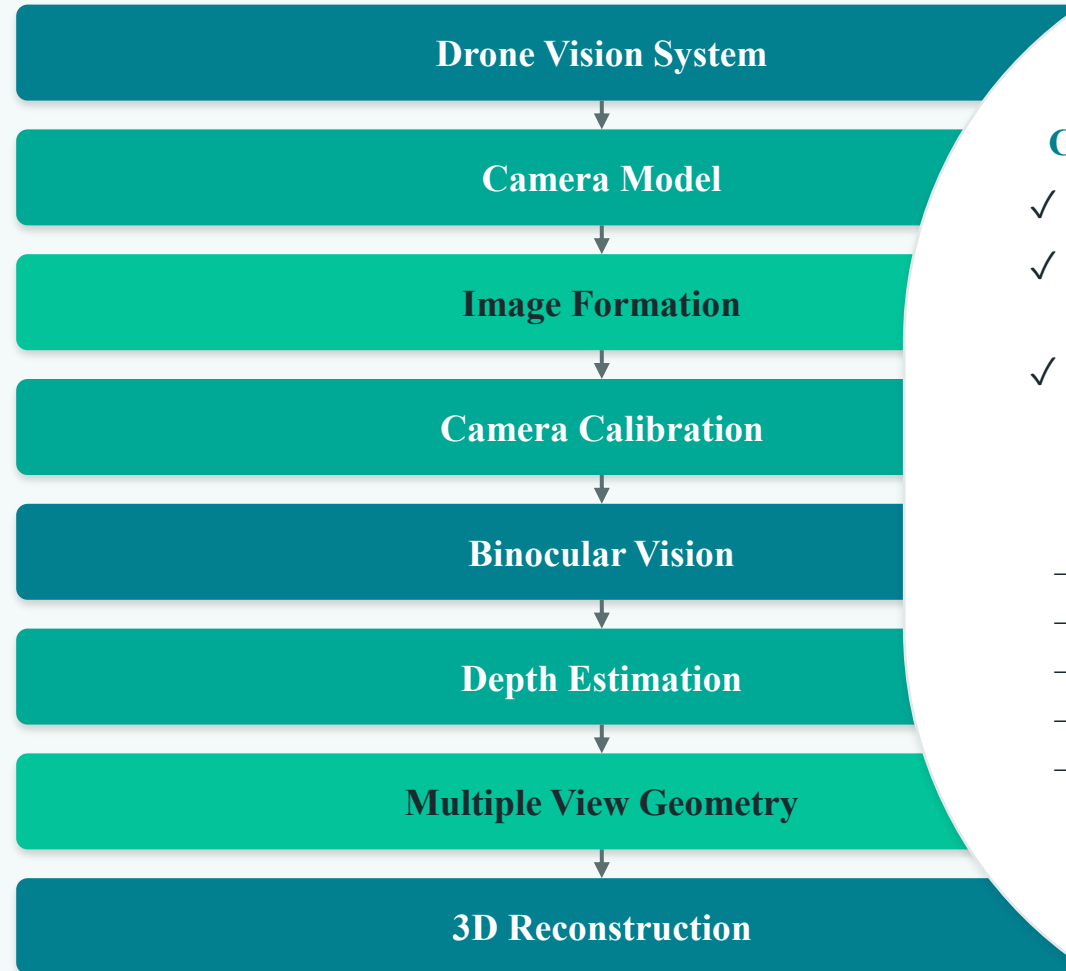


CHARACTERISTICS

- ✓ Introduces calibration as a step
- ✓ Concepts flow in a simple straight line
- ✓ Limited flow — no branching or feedback
- ✓ Weak linking between concepts overall



Drone Vision System — Organized Pipeline



CHARACTERISTICS

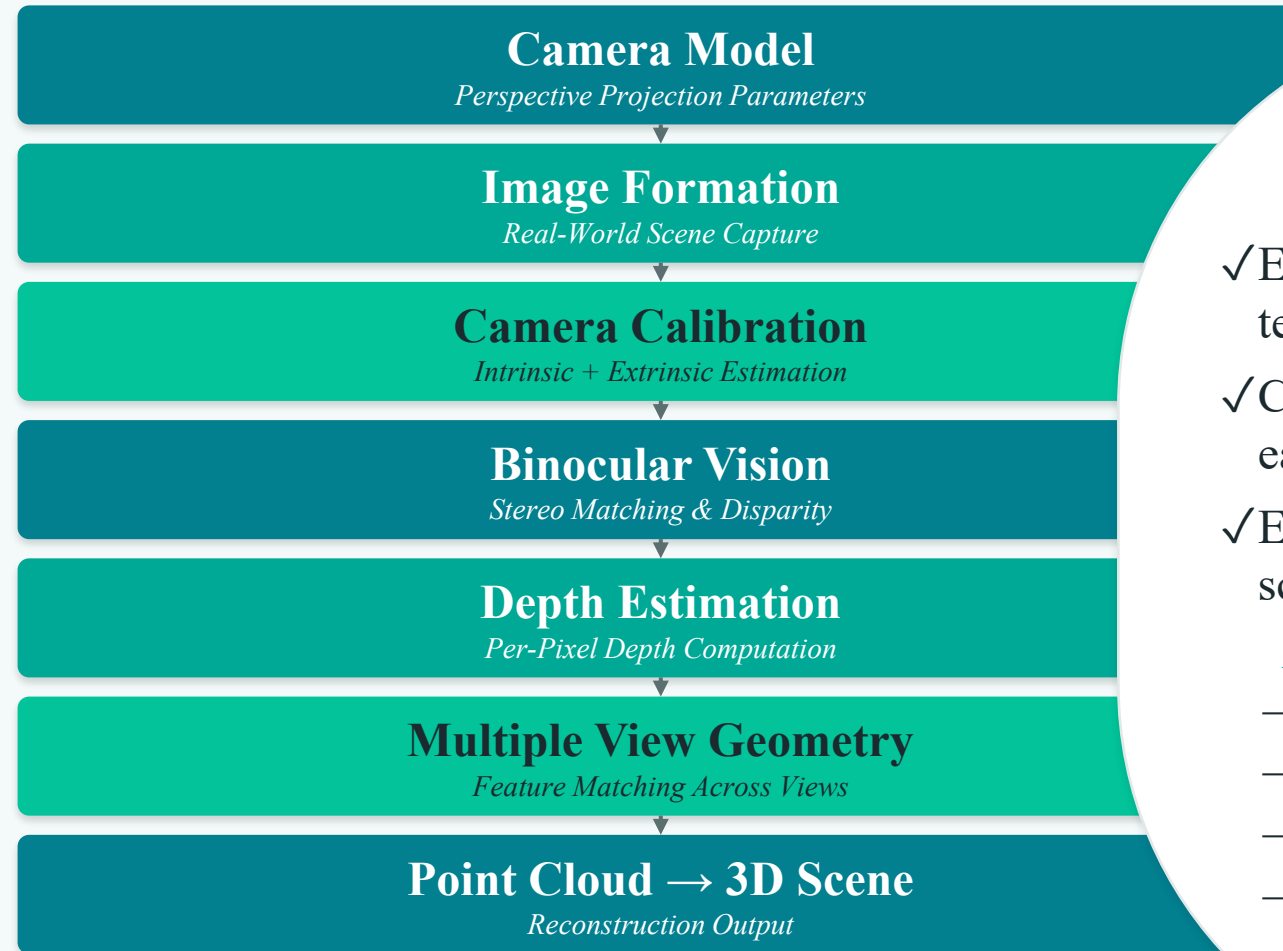
- ✓ Full pipeline from capture to 3D model
- ✓ Camera calibration and stereo vision included
- ✓ Multiple-view geometry ties depth data together

FLOW

- Capture
- Calibration
- Stereo Matching
- Depth
- 3D Model



Fully Integrated Reconstruction Pipeline



CHARACTERISTICS

- ✓ Every stage carries its own technical sub-detail
- ✓ Clear dependency chain — each stage feeds the next
- ✓ Ends in a validated 3D scene reconstruction

APPLICATIONS

- Autonomous Navigation
- Construction Survey
- Agriculture Monitoring
- Disaster Mapping
- Smart City Planning